

Midterm Exam

Version A

Friday, October 9, 2026 · in class

Name: _____ Student ID: _____

40 points | Suggested working time 40 minutes

Use a basic calculator. Show your working. Allow about 21 minutes for Part 1, 7 for Part 2, 6 for Part 3 and 3 for the Summary, plus 3 minutes to read and check. Part 1 and the Summary are marked on clear reasoning and clear explanations in your own words. The data are synthetic teaching data from a fictional seaside café.

Part	What it covers	Points	Score
1	Concepts: model choices, interpreting coefficients, outliers	24	
2	Calculations by hand: the t-test	5	
3	Computations in Sheets: functions and interpretation	8	
Summary	Summary for the owner	3	
Total		40	

PART 1 · CONCEPTS: model choices and interpretation

24 points · about 21 minutes

Q1 · Choose the inputs (7 points, about 6 minutes)

A seaside café wants to predict **tomorrow's gelato sales** (cups) **before it opens**. Table 1 lists information it could use. Across 168 days of data, **temperature and sunshine are strongly related to each other ($r = 0.78$)**. Their correlations with gelato sales are 0.58 (temperature) and 0.49 (sunshine).

Table 1. Candidate information.

#	Information
1	Tomorrow's temperature forecast (°C)
2	Tomorrow's sunshine forecast (hours)
3	Whether tomorrow is a weekend (yes or no)
4	Whether a beach event is on the calendar tomorrow (yes or no)
5	Tomorrow's planned gelato price (€)
6	Tomorrow's iced coffee sales (cups), counted at closing time

1a Choose **three** inputs for the model. For each, say whether it should push gelato sales **up or down**, and give a one-line reason. 3 pts

Input (name or number)	Up or down?	Reason (one line)

1b One candidate in Table 1 is **unsuitable** for this forecast. Which one, and what is the problem? 2 pts

1c Should temperature **and** sunshine both go into the same model? Explain, and say which one you would keep. 2 pts

Q2 · Interpret the coefficients (14 points, about 12 minutes)

A model of daily gelato sales (cups) was fitted to 168 days with $\text{=LINEST}(y, X, \text{TRUE}, \text{TRUE})$. X_1 = **temperature (°C)**, X_2 = **beach event (1 = event, 0 = no event)**, X_3 = **gelato price (€)**. In the data, temperature ran from 13 to 32 °C and the gelato price from €3.70 to €4.80. Table 2 shows how LINEST lays out its results; Table 3 is the output.

Table 2. How LINEST lays out its results.

Col A (X_3)	Col B (X_2)	Col C (X_1)	Col D (Intercept)	Statistic description
Slope m_3	Slope m_2	Slope m_1	Intercept (b)	Coefficients for: $y = m_1X_1 + m_2X_2 + m_3X_3 + b$
SE – m_3	SE – m_2	SE – m_1	SE – Intercept	Standard error for each coefficient
R^2	SE – Y	Blank	Blank	Quality of the model fit
F-statistic	df	Blank	Blank	Analysis of variance metrics
SS – Reg	SS – Resid	Blank	Blank	Sum of squares metrics

Table 3. LINEST output (shaded cells are blank).

Col A	Col B	Col C	Col D
–33.73	30.65	7.06	167.59
5.65	4.76	0.63	22.05
0.53	24.83		
61.31	164		
113418.35	101131.05		

2a Write the fitted equation, using the numbers in Table 3 and the variable names.

2 pts

2b (i) Interpret the **temperature** coefficient: what happens to predicted gelato sales when temperature rises by 1 °C, and what is held constant? (ii) In your own words, why do we say “holding the other variables constant”?

3 pts

2c (i) The **price** coefficient is negative. What does the sign mean for the café? (ii) The **event** coefficient is positive and event is a 0/1 variable. Interpret it in one sentence. (iii) A €1 price change is far bigger than the café would make (the price ranges only from €3.70 to €4.80). Rewrite the price coefficient for a **€0.10** increase.

3 pts

2d (i) Rewrite the temperature coefficient for a **10 °C** increase. (ii) Which is the better message for the owner, “per 1 °C” or “per 10 °C”? Why? (iii) A colleague says: “The event coefficient (30.65) is bigger than the temperature coefficient (7.06), so events matter more than temperature.” Is that a fair comparison? Use your answer to (i).

3 pts

2e Tomorrow is forecast to be 26 °C, with a beach event and the planned price at €4.30. (i) Predict gelato sales. (ii) The owner is thinking of **raising the price from €4.30 to €4.50**. By how much would predicted sales change, everything else the same?

3 pts

Q3 · Outliers (3 points, about 3 minutes)

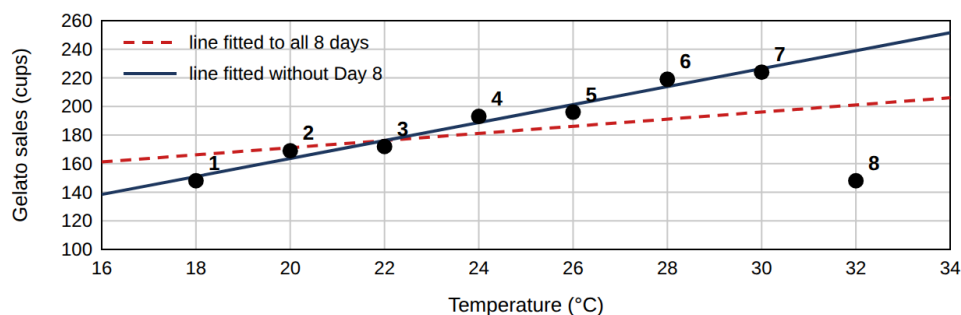
Constructed example: gelato sales on 8 days (Table 4, Figure 1) and the fitted line with and without Day 8 (Table 5).

Table 4. Temperature and sales on 8 days.

Day	1	2	3	4	5	6	7	8
Temperature (°C)	18	20	22	24	26	28	30	32
Sales (cups)	148	169	172	193	196	219	224	148

Table 5. Results for the straight line (sales against temperature).

	All 8 days	Without Day 8
Slope (cups per °C)	2.49	6.29
R ²	0.17	0.97
SE – Y (cups)	28.77	5.18

Figure 1. The 8 days, with the line fitted to all of them (dashed) and the line fitted without Day 8 (solid).

3a Which day does not follow the pattern of the others?

1 pt

3b What impact does that one day have on the slope and on R²?

1 pt

3c Before deleting it, what would you do?

1 pt

PART 2 · CALCULATIONS BY HAND: the t-test

5 points · about 7 minutes

Q4. On 30 June the café raised the gelato price from €4.00 to €4.50. Table 6 shows gelato sales (cups) on 5 weekdays **before** (Group 1) and 5 **after** (Group 2). Both samples are weekdays with similar weather (about 24 °C). Formulas as given in class:

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}} \quad s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \quad t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

4a Complete Table 6: the two means, the two sums of squared deviations, and s for each group (2 decimals).

3 pts

Table 6. Worksheet (fill in the blank cells).

Day	Before (x ₁)	x ₁ – $\bar{x}_1$	(x ₁ – $\bar{x}_1$) ²	After (x ₂)	x ₂ – $\bar{x}_2$	(x ₂ – $\bar{x}_2$) ²
1	196			184		
2	204			192		
3	188			176		
4	212			196		
5	200			182		
Sum						
Mean $\bar{x}$						
s						

4b Calculate s_p and t (2 decimals). A negative t only means that Group 2 has the larger mean.

2 pts

PART 3 · COMPUTATIONS IN SHEETS: functions and interpretation

8 points · about 6 minutes

Q5. On the sheet “Price test”, the “before” sales are in B2:B6 and the “after” sales in C2:C6 (Part 2). On the sheet “Daily data”, gelato sales are in B2:B169 and the three X columns in C2:E169 (Part 1).

- 5a** Write the Sheets formula for each (any correct form is accepted): (i) the sample standard deviation of the “before” sales; (ii) the p-value comparing “before” with “after”, two-tailed, equal variances; (iii) the regression with all its statistics (the LINEST output). 3 pts

- 5b** In `=T.TEST(B2:B6, C2:C6, 2, 2)`, what do the last two numbers mean? In `=LINEST(..., TRUE, TRUE)`, what does the second TRUE add to the output? 2 pts

Sheets gives `=T.TEST(B2:B6, C2:C6, 2, 2) = 0.031`.

- 5c** (i) State the null hypothesis in words. (ii) Is the difference statistically significant at 5%? Explain using the p-value. (iii) In plain words, what does $p = 0.031$ mean? 3 pts

SUMMARY FOR THE OWNER

3 points · about 3 minutes

- 6** Write 3–4 sentences to the café owner, who has no statistics background. Include: (i) one input from the Part 1 model and what a sensible change in it does to gelato sales, holding the others constant; (ii) what the price test showed; (iii) one caution. 3 pts

— End of exam —